GRAD-CAM Technology Overview

Gradient-weighted Class Activation Mapping | Explainable MRI classification

Project: Privacy-Preserving Federated Learning Framework for Early Brain Tumor Classification Using Multi-Hospital MRI Images

Technical overview | Research and project-evaluation reference

1. Introduction and purpose

Grad-CAM is a post-hoc explainability technique for convolutional neural networks. It produces a coarse spatial heatmap showing which image regions most influenced a selected class score. For MRI tumor classification, it helps investigators inspect model attention, but it does not diagnose disease or delineate tumor boundaries.

Original reference: Selvaraju et al., “Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization,” ICCV 2017.

2. Working principle

- Step 1: Preprocess an MRI image with the same resizing and normalization used during training.
- Step 2: Run the trained classifier and choose the class logit to explain.
- Step 3: Capture feature maps from a selected convolutional layer and calculate gradients of the class logit with respect to those maps.
- Step 4: Average gradients spatially to produce channel importance weights.
- Step 5: Form a weighted feature-map sum, apply ReLU, resize to the MRI resolution, and overlay as a colored heatmap.

Core relation: alpha_k = spatial mean of d(score_class)/d(A_k). Heatmap = ReLU(sum_k alpha_k * A_k). This is a class-specific importance map, not a pixel-wise segmentation prediction.

3. Integration with EfficientNet-B0 + Vision Mamba + KAN

Use a late convolutional feature layer of EfficientNet-B0 as the Grad-CAM target. Backpropagate the selected class logit through the entire hybrid pipeline (EfficientNet-B0 → token projection → Vision Mamba → KAN head). This explains the convolutional features contributing to the final hybrid prediction.

If the Mamba tokenization destroys the original spatial arrangement or gradients do not propagate correctly, check feature-map shapes, hook placement, and whether the selected logit depends on the hooked activations.

4. Outputs for the project dashboard

Display	What it tells the evaluator
Original MRI	Unaltered input image
Prediction and probabilities	Model-estimated tumor class and calibrated scores
Grad-CAM heatmap	Areas influencing the chosen class
Overlay	Visual comparison with MRI anatomy
Uncertainty warning	Indicates a low-confidence prediction under a validation-set policy

5. Quantitative validation and limitations

- If matching expert masks exist: evaluate pointing-game accuracy and overlap metrics such as IoU or Dice on thresholded heatmaps; report the chosen threshold.
- Test explanation stability under mild, clinically plausible input perturbations; include sanity checks such as model-parameter randomization.
- Grad-CAM can highlight background artifacts, scanners, or text overlays rather than tumors. A visually plausible heatmap does not prove clinical correctness.
- No tumor segmentation claim should be made without a separately validated segmentation model or appropriate expert masks.
- For external datasets, MRI sequence, label definition, preprocessing and patient independence must be compatible with the evaluation.

6. Example Colab implementation outline

Install torch, torchvision, timm, matplotlib and opencv-python. Register forward and backward hooks on the last spatial convolution block, save activation and gradient tensors, calculate channel-weighted heatmaps, normalize to [0,1], resize and blend with the original MRI. Clear hooks after each inference. Use model.eval() and enable gradients for Grad-CAM even when no training occurs.

7. Viva questions

- Why Grad-CAM? It makes the decision process more inspectable by highlighting influential regions.
- Grad-CAM versus segmentation? Explanations indicate influential regions; segmentation predicts region boundaries.
- Why not explain softmax probability? The raw class logit often provides more stable gradient signals.
- Can Grad-CAM establish clinical safety? No. It requires clinical evaluation and cannot independently verify diagnosis.

8. References

Selvaraju, R. R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV. https://arxiv.org/abs/1610.02391

Adebayo, J., et al. (2018). Sanity Checks for Saliency Maps. NeurIPS. https://arxiv.org/abs/1810.03292